Compiler CPCS302

2nd Semester 2021-2022 Mid Exam Female Dep.

This copy is reassembled by students of female department to help other students to practice exam for the subject.

The Mid-term exam covered the ideas of

Ch.1 – Ch.2 – Ch.3

May Allah bless you and good luck.

Solution Copy

King Abdulaziz University

CPCS302 – Midterm Exam

Student Name :.....................................

Student ID: ....

Student Section: ....

Grade out of 25	

Q1) Answer the following MCQ by choosing the right answer.

1- What is true about the interpreter among the following choices?
   a- Interpreter is slower than compiler
   b- It converts the source code into machine code
   c- Direct execute the operations specified in the source code
   d- Hard and less efficient in debugging

2- Tokens are the result of the parser, what of the following is incorrect to consider as a token? (I'm not sure 100% but like 70%)
   a- Keywords
   b- ?
   c- String of characters
   d- Integers

3- If we have set D = {0,1,2,….,9}, the regular expression that represent writing one D or more is
   a- D
   b- D*
   c- D^+
   d- DD^+

4- If we have a programming language that is working with different platforms, the programming language has: (The red is wrong for sure, but I don't recall which one is correct between "a" and "c", see Ch.1)
   a- One backend and multi-frontends
   b- Multi-frontends and multi-backends
   c- One frontend and multi-backends
   d- Multi-frontend and one middle end

5- In the concrete syntax tree, the intermediate node is
   a- Operand
   b- Operator
   c- Non termina
   d- Terminal

6- The **r** that is used in the regular expression **r?** produces
   a- Zero or more value of r
   b- One or more value of r
   c- Zero or one value of r
   d- ?

7- The effect of lexical error recovery might create syntax errors if handled by the Scanner (I'm not sure 100% but like 70%)
   a- True
   b- False

8- If there was an error of a lexeme, the error shows in the Scanner
   a- True
   b- False

9- The postfix of 8-3+5 is 835-+
   a- True
   b- False

10- If you want to produce a binary pattern of zero's and one's that always ends with four zeros, then the regular expression for this is (0|1)0000
   a- True
   b- False

Q2) Answer the following question

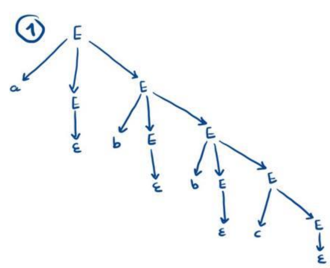
a- Is the following grammar ambiguous or not?

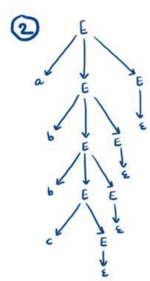
E→aEE
E→bEE
E→cE
E→ ε

1- Ambiguous
2- Unambiguous

if more than one tree can be produced, then its ambiguous

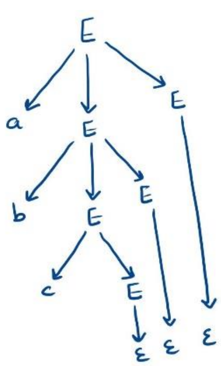
* Try "abbc"

1


2


∴ Ambiguous

b- Derive a parse tree using left derivation to produce this output: *abc*



Q3) This question shows a grammar of some programming language, answer the following questions.

$$\boxed{\begin{array}{l} S \rightarrow tTF \mid \varepsilon \\ T \rightarrow fT \mid \varepsilon \\ F \rightarrow hF \mid \varepsilon \end{array}} \longleftarrow R$$

a- Determine the four tuples of the grammar above

R = The whole grammar (the box)

S = {S}

V= {S, T, F}

∑ = {t, f, h}

b- Use the **right** derivation to produce this output: *tfh*

S → tTF (The start symbol, we start from the beginning of the grammer)

→ tThF (We replaced the "F" from step 1 into "hF")

→ tTh (We replaced the "F" from step 2 into "ε")

→ tfTh (We replaced the "T" from step 3 into "fT")

→ tfh (We replaced the "T" from step 4 into "ε")

Q4) The following grammar is ambiguous, answer the question about this grammar:

E → E * E | E / E | D

D → 0 | 1 | 2 |…..| 9

a- What is the reason behind the ambiguity?

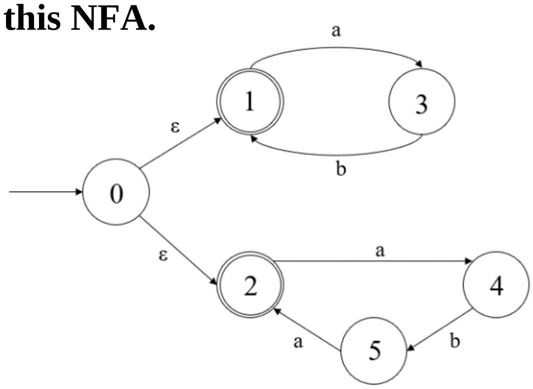
Left associative

b- Fix the ambiguity.

E → E * D | E / D | D

D → 0 | 1 | 2 |…..| 9

Q5) The following graph represent a NFA of a certain regular expression, answer the following question using this NFA.



a- Find the following using the above NFA

Find explanation at CH.3, slides 53 - 55

F: {1 , 2} (F represent every final state, in this example they are states 1 and 2)

Σ: {a , b} (∑ represent the inputs, in this example they are a and b)

b- Fill the following transition table using the above NFA

State	a	b	ε
0	ϕ	ϕ	{1,2}
1	{3}	ϕ	ϕ
2	{4}	ϕ	ϕ
3	ϕ	{1}	ϕ
4	ϕ	{5}	ϕ
5	{2}	ϕ	ϕ

c- Check if the following string and answer the questions.

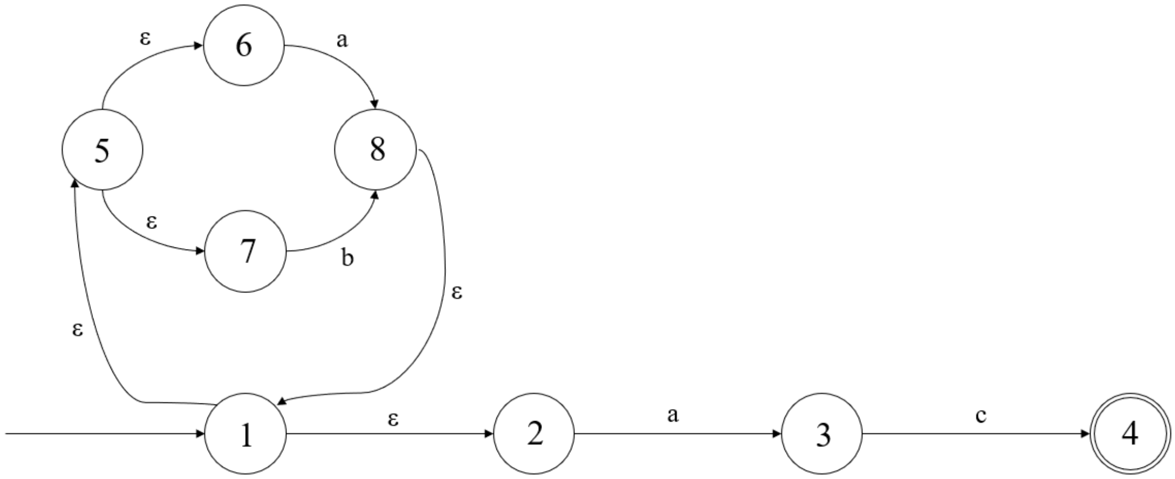
1- The pattern *abaa* is (Accepted/Rejected) using the above NFA. Draw the path:

$0 \xrightarrow{\varepsilon} 2 \xrightarrow{a} 4 \xrightarrow{b} 5 \xrightarrow{a} 2 \xrightarrow{a} 4$

2- The pattern *abab* is (Accepted/Rejected) using the above NFA. Draw the path:

$0 \xrightarrow{\varepsilon} 1 \xrightarrow{a} 3 \xrightarrow{b} 1 \xrightarrow{a} 3 \xrightarrow{a} 1$

Q6) Answer the following questions using this NFA



a- Find the epsilon closure for state {1}:
ε-closure ({1}): {1,2,5,6,7}

b- Find the transition from state 1 to the symbol a:
Move (ε-closure ({1}) , a): => Move({1,2,5,6,7} , a) => {3,8}

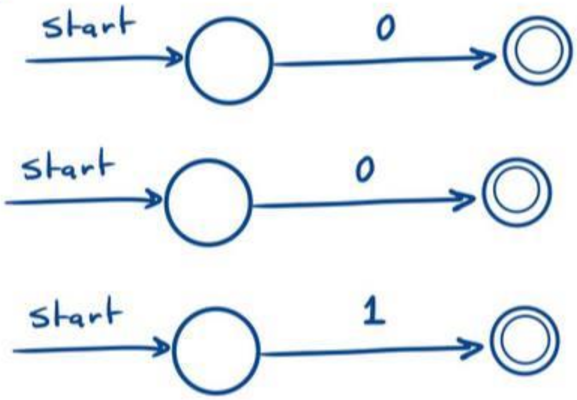
c- Find the transition from the state 1 to the symbol b:
Move (ε-closure ({1}) , b):): => Move({1,2,5,6,7} , b) => {7}

Q7) for the following regular expression: 0* | (10)

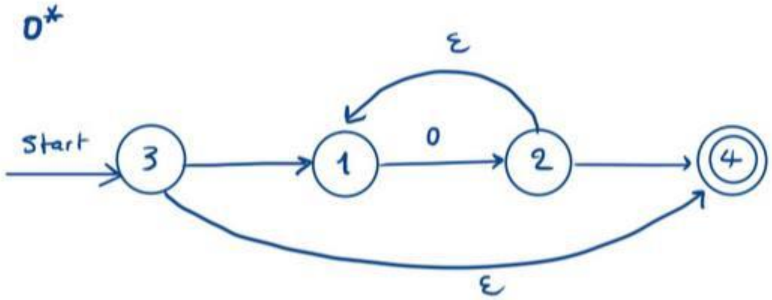
<u>Note: the (10) is not ten, it's 1 concatenation 0.</u>

Draw solution here:

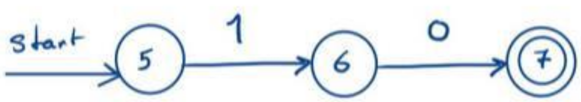
1)



2) 0^*



3) (10)



4) $0^* | (10)$